Problem 1a) (15 points)

Given the image region as shown in Figure 1 and $V = \{1\}$, what is the shortest m-path between p (the pixel at the upper-left corner) and q (the pixel at the bottom-right corner)? Give the length of the path if the path exists.

$$\begin{array}{ccccc}
1 & 0 & 1 & 1 & 0 \\
0 & 1 & 0 & 1 & 0 \\
0 & 0 & 1 & 1 & 0 \\
1 & 1 & 0 & 1 & 0 \\
0 & 1 & 1 & 1 & 1
\end{array}$$

Step 1: Identify pixels in V

p

1	0	1	1	0
0	1	0	1	0
0	0	1	1	0
1	1	0	1	0
0	1	1	1	1

q

All pixels with value 1 are highlighted. These are the only candidates for the m-path since $V = \{1\}$.

Step 2: Apply m-adjacency rules

Two pixels with values in V are **m-adjacent** if:

a) They are 4-adjacent (share an edge), OR

b) They are diagonally adjacent AND their shared 4-neighbors contain no pixels with values in V.

The second condition prevents diagonal shortcuts when a 4-connected bridge exists, eliminating path ambiguity.

Step 3: Trace the shortest m-path

p

1	0	1	1	0
0	1	0	1	0
0	0	1	1	0
1	1	0	1	0
0	1	1	1	1

q

Path: $(0,0) \to (1,1) \to (2,2) \to (2,3) \to (3,3) \to (4,3) \to (4,4)$

Diagonal move justifications:

- $(0,0) \to (1,1)$: Shared 4-neighbors are $(0,1) = 0$ and $(1,0) = 0$. Neither is in V, so m-adjacent.
- $(1,1) \to (2,2)$: Shared 4-neighbors are $(1,2) = 0$ and $(2,1) = 0$. Neither is in V, so m-adjacent.

Results

The shortest m-path exists and has **length 6**.

Problem 1b) (15 points)

To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain.

Step 1: Understand the Scene Characteristics

Fireworks at night present specific challenges:

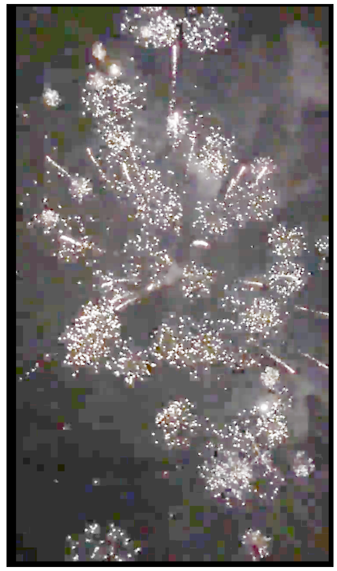
- **High dynamic range**: Very bright fireworks against a very dark sky
- **Desired appearance**: Pure black sky with vivid, crisp firework sparks
- **No hidden information**: The dark sky contains nothing meaningful to reveal

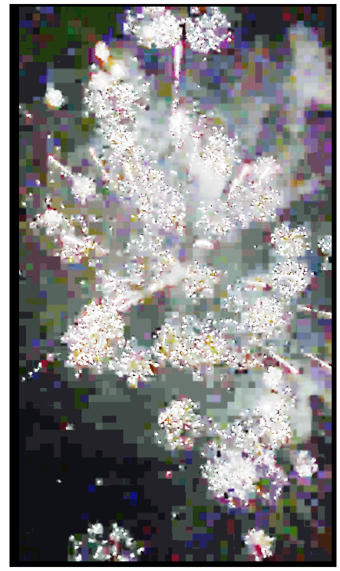
Original fireworks image

Step 2: Brightening Operations

These operations attempt to reveal detail in shadows by lifting dark intensities.


Gamma ($\gamma = 0.5$)

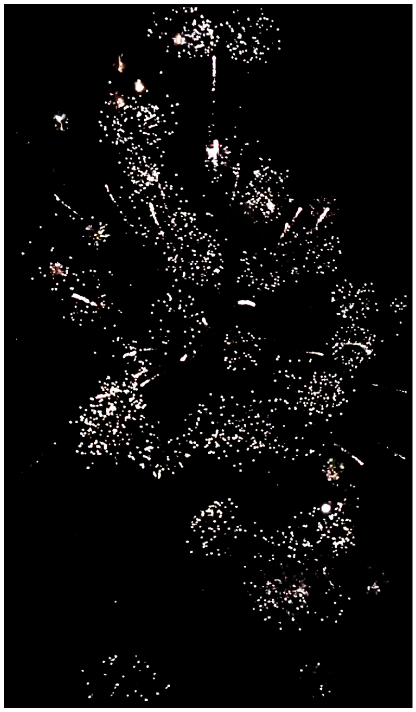

Log Transform


Histogram Equalization

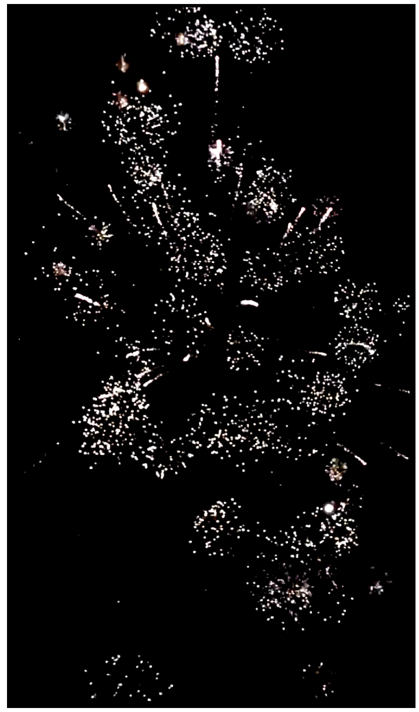
Effect: The black sky becomes gray, revealing noise and sensor artifacts. This is **undesirable** for fireworks because the sky *should* be black—there is no meaningful information to reveal.

Step 3: Contrast Enhancement Operations

These operations push dark values toward black and light values toward white.



S-Curve Contrast



Boosted Contrast

Effect: The sky becomes pure black while the fireworks remain bright. This is **desirable** for aesthetic night photography.

Step 4: Sharpening Operations

These operations enhance edges and fine details.

Unsharp Masking

High-Boost Filtering

Effect: Firework sparks appear crisper with more defined edges.

Step 5: Color Enhancement Operations

Saturation Boost

Warm Temperature

Warm + Color Contrast

Saturation: Uniformly boosts all color intensity—can cause clipping in already-vivid areas.
Warm Temperature: Shifts colors toward orange/yellow by boosting red and reducing blue channels, enhancing the fiery appearance.
Warm + Color Contrast: Applies warmth first, then an S-curve in Lab color space to push warm colors warmer and cool colors cooler, creating rich golden tones.

Step 6: Consider the Optimization Goal

The "correct" operations depend entirely on what you are trying to achieve:

Goal	Recommended Operations	Avoid
Aesthetic (fireworks)	S-curve, sharpening, saturation	Gamma, Hist EQ
Surveillance	Gamma, log transform, Hist EQ	S-curve
Medical imaging	CLAHE, careful contrast	Extreme transforms
Object detection	Thresholding, edge detection	Aesthetic enhancements

The same operation can be *beneficial* or *harmful* depending on the goal. Histogram equalization is excellent for revealing hidden details in surveillance footage, but terrible for fireworks photography where the dark background should remain dark.

Step 7: Consider the Display Medium

The target display device also affects which operations are most beneficial:

Display	Prioritize	Less Important
Mobile phone (small screen)	Color, contrast, brightness	Fine sharpening
Desktop monitor	Sharpness, fine detail	Can be subtle
Print	Resolution, color accuracy	Different gamma curve

On a mobile phone, viewers cannot resolve fine detail, so aggressive sharpening provides little benefit—but punchy colors and contrast help images "pop" on a small screen viewed in variable lighting. On a desktop monitor, fine details are visible, making sharpening more valuable.

Results

Recommended operations for fireworks photography:

a) **S-curve contrast enhancement**: Apply a sigmoid transformation $s = \frac{1}{1+e^{-k(r-0.5)}}$ to push dark pixels darker and bright pixels brighter. This ensures a pure black sky while maintaining bright firework sparks.

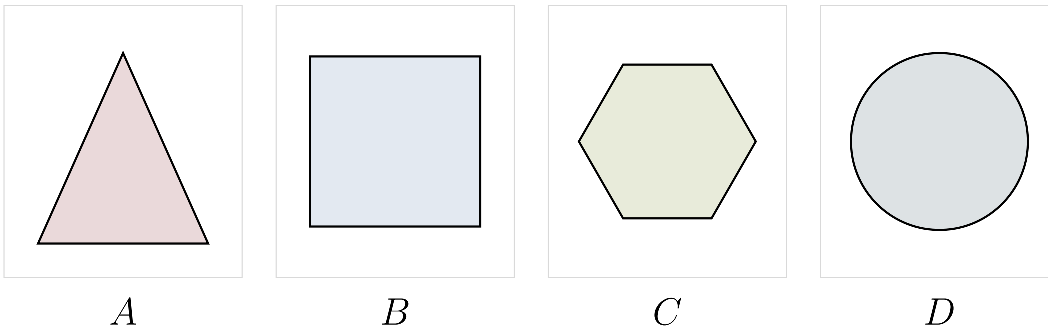
b) **Saturation boost** (especially for mobile display): Increase color saturation to make the firework colors more vivid and visually striking.

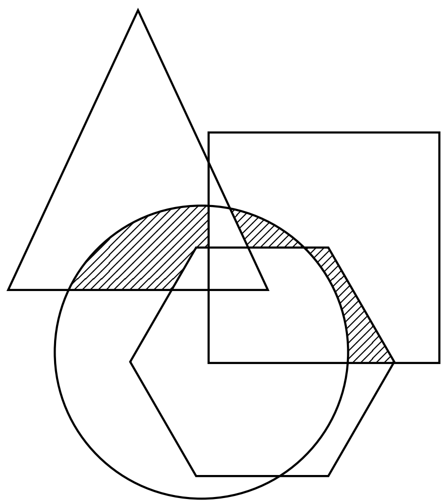
c) **Sharpening** (especially for desktop/print): Apply unsharp masking or high-boost filtering to enhance the crispness of firework spark edges.

Operations to avoid: Gamma correction ($\gamma < 1$), log transforms, and histogram equalization—these lift the dark sky and reveal noise, degrading the aesthetic quality.

Problem 2) (20 points)

A, B, C and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set





Step 1: Identify the Shaded Regions

Examining the diagram, the shaded area consists of three distinct regions where exactly two shapes overlap, excluding the areas where three or more shapes would overlap:

a) The region where A (triangle) and D (circle) overlap, but not B or C
b) The region where B (rectangle) and D (circle) overlap, but not A or C
c) The region where B (rectangle) and C (hexagon) overlap, but not A or D

Step 2: Write Set Expressions for Each Region

Using set intersection ($\cap$) and set difference ($\setminus$ or complement notation):

Region 1: Points in both A and D, but not in B or C:

$$(A \cap D) \setminus (B \cup C) = A \cap D \cap B^c \cap C^c$$

Region 2: Points in both B and D, but not in A or C:

$$(B \cap D) \setminus (A \cup C) = B \cap D \cap A^c \cap C^c$$

Region 3: Points in both B and C, but not in A or D:

$$(B \cap C) \setminus (A \cup D) = B \cap C \cap A^c \cap D^c$$

Step 3: Combine Using Union

The complete shaded region is the union of all three regions.

Results

The shaded set can be expressed as:

$$\boxed{[(A \cap D) \setminus (B \cup C)] \cup [(B \cap D) \setminus (A \cup C)] \cup [(B \cap C) \setminus (A \cup D)]}$$

Or equivalently, using complement notation:

$$(A \cap D \cap B^c \cap C^c) \cup (B \cap D \cap A^c \cap C^c) \cup (B \cap C \cap A^c \cap D^c)$$

Problem 3) (25 points)

Given a 3x3 spatial filter $\frac{1}{9}\begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ answer the following questions:

a) (10 points) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? Why? (Hint: is this filter an image smoothing filter, an image sharpening filter or an edge detector?)

b) (15 points) Perform the image convolution using this filter on a 3x3 image $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$.

Give the CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping)

Problem 4) (25 points)

An image with intensities in the range $r \in [0, L-1]$ has the pdf

$$p_r(r) = \begin{cases} \dfrac{4r}{3(L-1)^2} & 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

where $L-1$ is the maximum intensity value. It is desired to transform the intensity levels of this image so that they will have the specified pdf

$$p_z(z) = \begin{cases} \dfrac{2z}{(L-1)^2} & 0 \leq z \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

Assume that the intensity value is continuous and find the transformation function $z = T(r)$ in terms of r and z that will accomplish this.

(Hints: The transformation function of histogram equalization is given as

$$s = T(r) = (L-1)\int_0^r p_r(w)\,dw$$

)

Bonus Question (20 points)

Perform the Fourier transform to a function as shown in Figure 3.

(Hint: the definition of the Fourier transform of a function is $F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut}\, dt$)